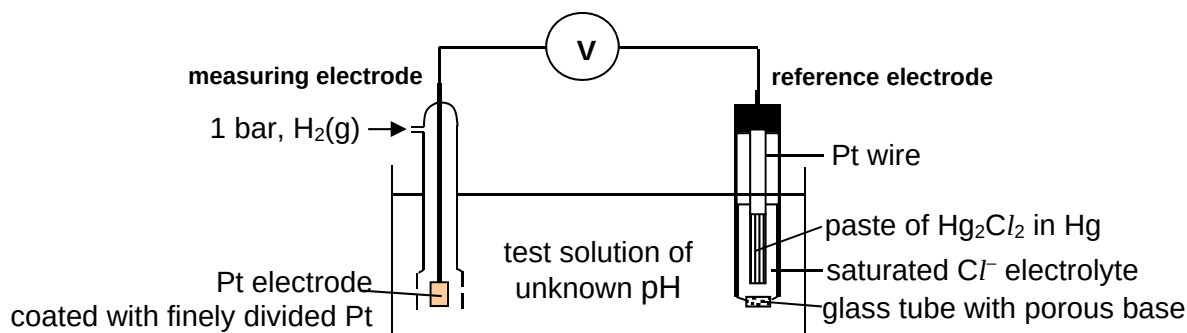


Section A (60 marks)

Answer **all** the questions in this section on the writing papers provided.

- 1 (a) A student designed an ion-specific probe to determine the concentration of H^+ ions in solution. He connected a hydrogen electrode to a calomel reference electrode, which comprises a paste of mercury(I) chloride, Hg_2Cl_2 , and mercury in a saturated chloride electrolyte. The reference electrode is inserted into a glass tube with a porous base.



The hydrogen electrode, which is sensitive to H^+ ion concentration, is the measuring electrode. Therefore, this galvanic cell acts as a pH meter.

When both electrodes are in contact with the test solution, an electric current flows through the wire and mercury(I) chloride reacts to form liquid mercury and chloride ion at the reference electrode.

Under standard conditions, the $E^\ominus_{\text{cell}}$ of the pH meter is +0.28 V.

- (i) Suggest the use of the glass tube with a porous base in this experimental set-up. [1]

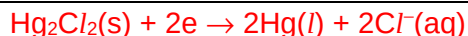
To act as a salt bridge.

- (ii) Identify the half-cell containing the anode. [1]

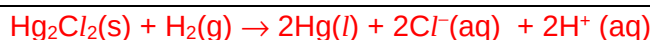
The half-cell with the hydrogen electrode contains the anode.

In the half-cell containing the calomel reference electrode, Hg^+ undergoes reduction to form Hg. Hence it contains the cathode.

- (iii) Construct the half equation for the reaction occurring at the reference electrode. [1]



- (iv) Hence write an overall balanced equation for the pH meter. [1]



- (v) Determine the $\Delta G^\ominus$ of this reaction and state the units.

[2]

$$\Delta G^\ominus = -nFE^\ominus_{\text{cell}}$$

$$\Delta G = -(2)(96500)(+0.28) = -54000 \text{ J mol}^{-1} \text{ or } -54.0 \text{ kJ mol}^{-1}$$

- (vi) Explain why the cell potential would increase when the pH meter is placed in a solution of a higher pH.

[1]

At a higher pH, there is lower $[H^+]$,

- equilibrium position in $2H^+(aq) + 2e \rightleftharpoons H_2(g)$ shifts to the left (OR eqm shifts to favour oxidation of H_2) and thus
- $E(H^+/H_2)$ becomes (more) negative.

Thus E_{cell} becomes more positive.

- (b) The student used his probe in (a) to analyse a sample of mandelic acid, a bitter almond extract that is used as an antibacterial treatment.

The student obtained a cell potential reading, E_{cell} , of +0.41 V.

The *Nernst equation* can be used to calculate the concentration of H^+ ions detected by the probe:

$$E_{\text{cell}} = E^\ominus_{\text{cell}} - \frac{0.0592}{n} \log_{10} \left(\frac{[H^+]^2 [Cl^-]^2}{P_{H_2}} \right)$$

where n is the number of moles of electrons transferred in the overall reaction,
 $[H^+]$ is the concentration of hydrogen ions detected,
 $[Cl^-]$ is the concentration of Cl^- ions,
 P_{H_2} is the pressure, in bar, of hydrogen gas passed into the probe.

Assuming that the concentration of Cl^- is constant at 1 mol dm^{-3} , use the *Nernst equation* to calculate the concentration of H^+ ions in the solution and the pH of the solution.

[2]

$$E = E^\ominus_{\text{cell}} - \frac{0.0592}{n} \log \left(\frac{[H^+]^2 [Cl^-]^2}{P_{H_2}} \right)$$

$$0.41 = 0.28 - \frac{0.0592}{2} \log([H^+]^2)$$

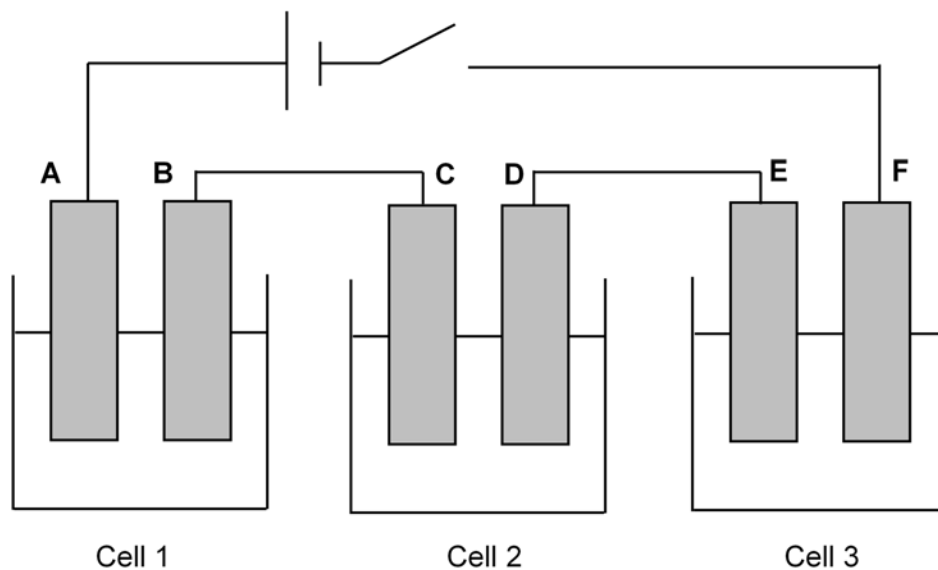
$$0.41 = 0.28 - \frac{0.0592}{2} \times 2 \log[H^+]$$

$$\log[H^+] = -\frac{0.13}{0.0592}$$

$$[H^+] = 6.37 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log(6.37 \times 10^{-3}) = 2.20$$

- (c) A current is passed through three cells connected in series as shown below.



Cell 1 contains lead electrodes, **A**, **B** and electrolyte $\text{Pb}(\text{NO}_3)_2(\text{aq})$.

Cell 2 contains platinum electrodes, **C**, **D** and electrolyte $\text{NaBr}(\text{aq})$.

Cell 3 contains platinum electrodes, **E**, **F** and electrolyte containing $\text{MnCl}_x(\text{aq})$.

- (i) On closing the switch, write equations, including state symbols, for the reactions occurring at electrodes **A** and **D**.

[2]



- (ii) A current of 0.36 A was passed through the circuit for 30 minutes. 0.183 g of manganese was deposited in Cell 3. Determine the value of x in MnCl_x .

[3]

$$Q = 0.36 \times 30 \times 60 = 648\text{C}$$

$$\text{Amount of electrons transferred} = 648/96500 = 0.0067150 \text{ mol}$$

$$\text{Amount of Mn deposited} = 0.183/54.9 = 0.0033333 \text{ mol}$$

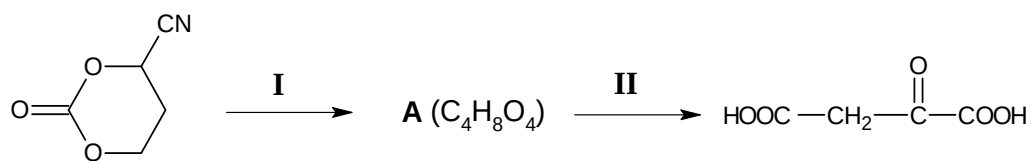


$$\frac{n_e}{n_{\text{Mn}}} = \frac{0.0067150}{0.0033333} = 2$$

$$x=2$$

- (d) Identify the structures of compounds **A**, **B**, **C** and **D** and suggest the reagents and conditions for each numbered steps **I**, **II** and **III** where possible.

(i)



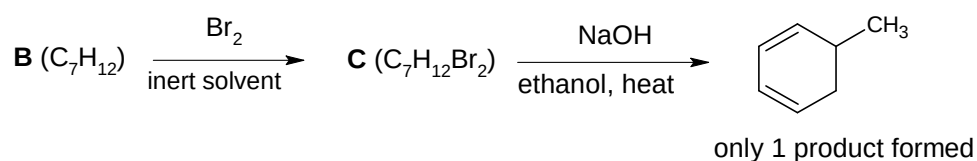
[3]

I: aqueous H_2SO_4 , heat

II: KMnO_4 (or $\text{K}_2\text{Cr}_2\text{O}_7$), aqueous H_2SO_4 , heat

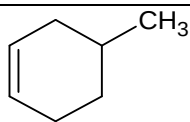
A: $\text{HOCH}_2\text{CH}_2\text{CHOH}(\text{COOH})$

(ii)

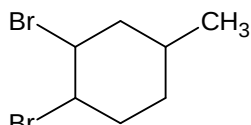


[2]

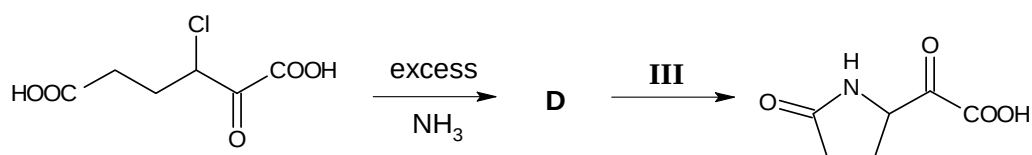
B:



C:

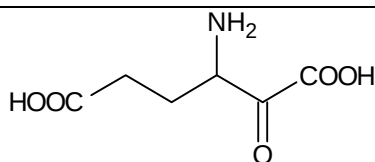


(iii)



[2]

D:



III: anhydrous PCl_5 (limited)

[Total: 21]

- 2 (a) Ethiopia is severely impacted by inadequate safe drinking water, where halogen contaminants such as chloride in drinking water, are the root cause of many common ailments such as low immunity and hypertension.

The Mohr's method is used to determine the chloride ion concentration of water samples from various sources such as seawater and stream.

A researcher retrieved a water sample from a stream and carried out Mohr's method, where the chloride ions present in the water sample is titrated against aqueous silver nitrate solution. As the aqueous silver nitrate is slowly added, a sparingly soluble white precipitate is formed. The indicator used in the titration is aqueous potassium chromate(VI), K_2CrO_4 . The end-point of the titration is reached when any excess silver nitrate added results in the formation of a red-brown precipitate of Ag_2CrO_4 .

The following are relevant K_{sp} values at 298 K:

silver salt	solubility product, K_{sp}
$AgCl$	2.00×10^{-10}
Ag_2CrO_4	1.10×10^{-12}

- (i) Determine the concentration of Cl^- (aq) ions at the end-point of the titration.

[1]

At the end point, all the Cl^- in the solution would be precipitated out as $AgCl$.

Let x be the solubility of $AgCl$.

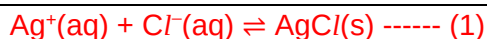
$$K_{sp} = [Ag^+][Cl^-]$$

$$2.00 \times 10^{-10} = x^2$$

$$x = 1.41 \times 10^{-5} \text{ mol dm}^{-3}$$

- (ii) Explain, with the aid of relevant equations, what would be observed if excess aqueous ammonia is added to the reaction mixture containing the white precipitate.

[2]



For $AgCl$, when ammonia is added, formation of soluble complex $Ag(NH_3)_2^+$ lowers the concentration of Ag^+ and causes position of equilibrium (1) to shift to the left. The ionic product $[Ag^+][Cl^-] < K_{sp}$ value and hence $AgCl$ dissolves.

- (iii) Determine the minimum concentration of CrO_4^{2-} (aq) ions in the titration mixture that is required to precipitate Ag_2CrO_4 immediately after the end-point.

[2]

$$K_{sp} = [Ag^+]^2[CrO_4^{2-}]$$

For precipitation of Ag_2CrO_4 just after end-point, ionic product = K_{sp}

$$(1.41 \times 10^{-5})^2 [CrO_4^{2-}] = 1.10 \times 10^{-12}$$

$$[CrO_4^{2-}] = 5.53 \times 10^{-3} \text{ mol dm}^{-3}$$

- (b) Oxygen gas, O_2 , reacts with various period 3 elements to produce compounds for numerous uses. For example, aluminium oxide, Al_2O_3 , is often used as a refractory material.

Element **X** forms a white oxide that is soluble in cold water. Its chloride dissolves in water to give a neutral solution.

Element **Y** forms an oxide which has the shape of trigonal planar around the central atom.

1 mole of the oxide of element **X** is added to an aqueous solution containing the same amount of the oxide of element **Y** to form a neutral solution.

Given that **X** and **Y** are period 3 elements, identify element **X** and the oxide of element **Y**. Give an equation to show the formation of the neutral solution.

[2]

X: Na **Oxide of Y: SO_3**



- (c) Reduction of ozone, O_3 , produces the ozonide anion, O_3^- . Derivatives of this anion are explosive and must be stored at very low temperatures.

Potassium hydroxide reacts with ozone to produce the corresponding metal ozonide, oxygen gas and water.

- (i) Suggest an equation for the reaction between potassium hydroxide and ozone.

[1]



Other balanced equations are also acceptable.

- (ii) Draw a dot-and-cross diagram for ozone. Using VSEPR theory, explain the shape and bond angle of ozone.

[3]



There is 1 lone pair and 2 bond pairs of electrons on the central atom. Electron pairs are arranged as far apart as possible to minimize repulsion Hence it adopts a bent shape.

Repulsion between lone pair-bond pair electrons is greater than the bond pair-bond pair electrons. Hence the bond angle is 117° (or 118°).

- (iii) All the O–O bonds in ozone are found to have identical bond lengths.

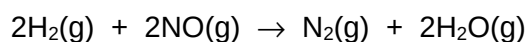
Suggest an explanation for this observation.

[1]

The pi electrons are delocalised over the 3 electronegative oxygen atoms.

(Any other phrasing related to resonance is acceptable.)

- (d) Hydrogen reacts with nitrogen monoxide to give nitrogen and steam as shown.



In order to determine the rate equation for this reaction, an investigation was carried out at a constant temperature and with the same concentration of NO for each experiment. The following results were obtained.

experiment	initial $[\text{H}_2]$ / mol dm^{-3}	initial rate of production of N_2 / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	1×10^{-3}	3×10^{-3}
2	2×10^{-3}	6×10^{-3}
3	3×10^{-3}	9×10^{-3}

- (i) Using the above data, deduce the order of reaction with respect to $[\text{H}_2]$. Hence sketch a graph of initial rate against $[\text{H}_2]$.

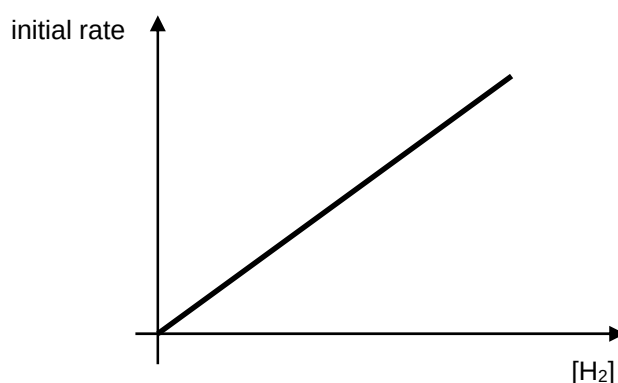
[2]

Let the rate equation be $\text{rate} = k [\text{H}_2]^x [\text{NO}]^y$

Comparing experiment 1 and 2,

$$\left(\frac{1 \times 10^{-3}}{2 \times 10^{-3}}\right)^x = \left(\frac{3 \times 10^{-3}}{6 \times 10^{-3}}\right)$$

$$x = 1$$



- (ii) The concentration of NO was halved and a new series of experiments were carried out at the same temperature. When a similar graph was plotted, the magnitude of the gradient decreases by 4 times compared to that obtained from the graph in (d)(i).

Deduce the order of reaction with respect to [NO].

[1]

When [NO] decreases by 2 times, rate decreases by 4 times.

Hence order of reaction wrt [NO] is 2.

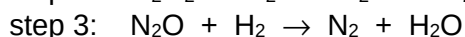
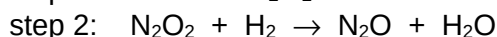
- (iii) Hence write the rate equation for the reaction between H₂ and NO. Include the units for the rate constant.

[1]

rate = $k [\text{NO}]^2 [\text{H}_2]$

units of $k = \text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$

- (iv) The following mechanism was proposed for the reaction of hydrogen with nitrogen monoxide.



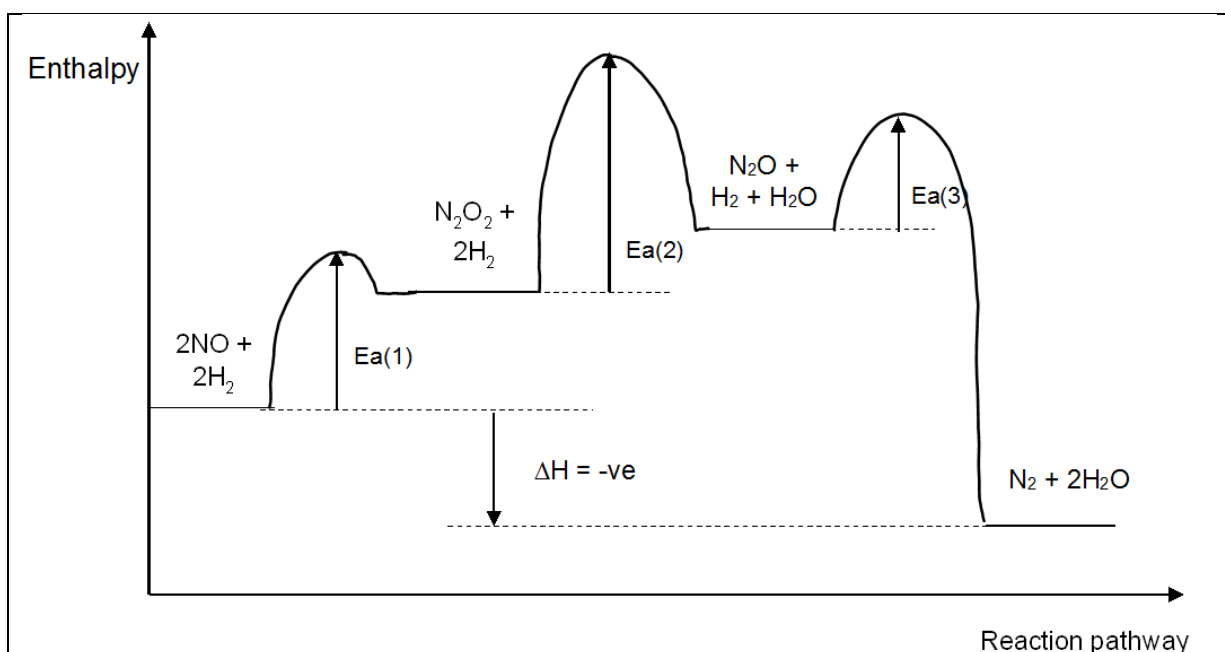
Identify the slow step which is consistent with the rate equation in (d)(iii).

[1]

The second step is the slow step.

- (v) Assuming that the reaction in (d) is exothermic, sketch a labelled energy profile diagram for the proposed mechanism in (d)(iv). Indicate clearly the activation energy for each step.

[3]



- ✓ Axes correctly labelled
- ✓ 3 humps
- ✓ Ea(2) highest hump
- ✓ ΔH correctly shown
- ✓ All correct reactants, intermediates and products

[Total: 20]

- 3 (a) The protein, somatostatin has been used mainly for its anti-secretory effects in gastrointestinal disorders. Recently, its effects in the treatment of cancer have also been reported.

- (i) Complete hydrolysis of a protein produces individual amino acids, but partial hydrolysis can break the protein down into dipeptide or tripeptide fragments.

State the reagents and conditions for the hydrolysis of a somatostatin molecule.

[1]

aqueous H_2SO_4 (or NaOH), heat under reflux

- (ii) A somatostatin polypeptide, containing **fourteen** amino acids, produced the following fragments on hydrolysis.

Ser- Cys
 Phe – Trp – Lys
 Cys – Lys – Asn
 Ala – Gly – Cys
 Asn – Phe – Phe
 Lys – Thr – Phe
 Phe – Thr – Ser

Using the same 3-letter abbreviations as above, deduce the amino acid sequence of the polypeptide.

[2]

Ala- Gly –Cys
 Cys – Lys – Asn
 Asn – Phe – Phe
 Phe – Trp – Lys
 Lys – Thr – Phe
 Phe – Thr – Ser
 Ser – Cys

Ala – Gly – Cys – Lys – Asn – Phe – Phe – Trp – Lys – Thr – Phe – Thr – Ser – Cys

- (iii) The complete hydrolysis of somatostatin is found to occur spontaneously at **high temperature**.

Suggest an explanation for this observation, taking into account the thermodynamic considerations of the reaction.

[2]

- Both ΔS and ΔH are positive
- At higher T, $\Delta G < 0$ as $|T\Delta S| > |\Delta H|$

- (b) Organophosphates, like carbon monoxide, are toxic to humans. When ingested, organophosphates inhibit the enzyme acetylcholinesterase, resulting in neuro poisoning. Carbon monoxide, on the other hand, inhibits the protein haemoglobin.

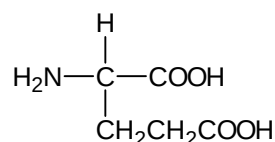
One possible treatment of neuro poisoning includes the use of a solution containing carbamates, with the general formula ROCONR_2 . The carbamate administered can stabilise the enzyme by forming an enzyme–carbamate complex, which stops the organophosphates from binding to the enzyme's active site.

Explain, in terms of ligand strength and the type of reaction occurring, why carbamate can be used to treat neuro poisoning.

[2]

- Carbamate is a stronger ligand than organophosphate.
- It undergoes ligand exchange reaction and binds more strongly via dative bond to the enzyme to form a more stable complex.
- Hence, it stops the organophosphates from binding to the enzyme's active site.

- (c) Glutamic acid is used as a flavour enhancer and is responsible for umami, one of the five basic tastes of human sense of taste. It has the structure of

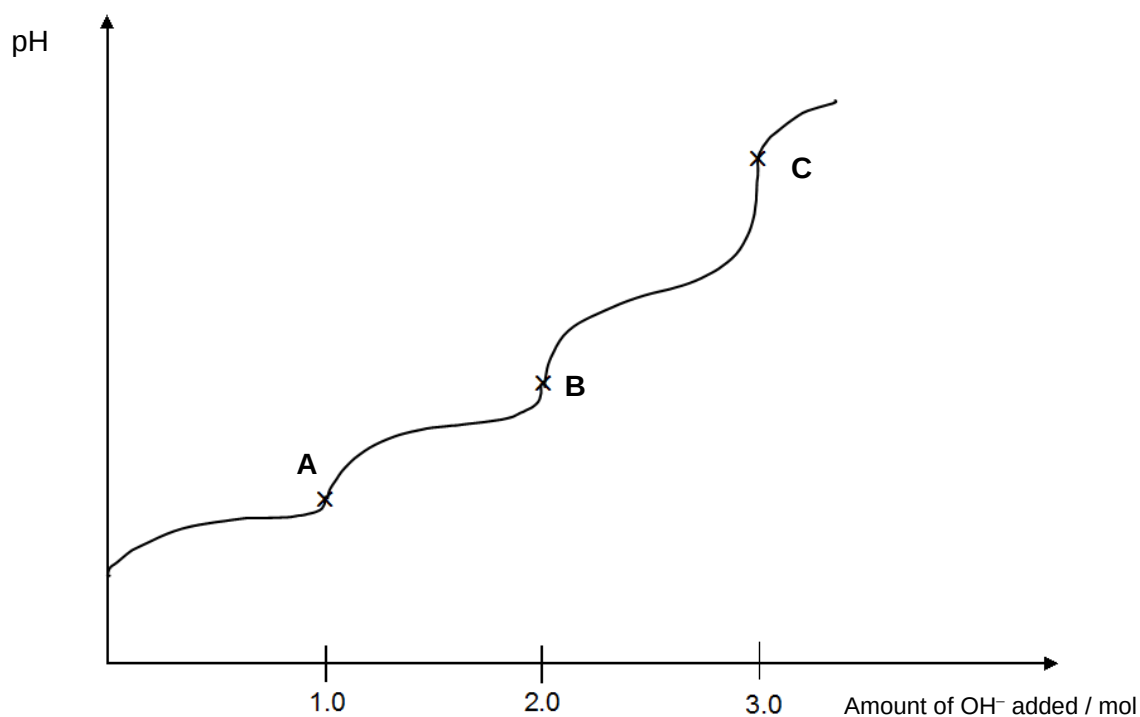


The $\text{p}K_{\text{a}}$ values and titration curve of 1 mole of the protonated form of glutamic acid are as shown.

$$\text{p}K_{\text{a}1} = 2.10$$

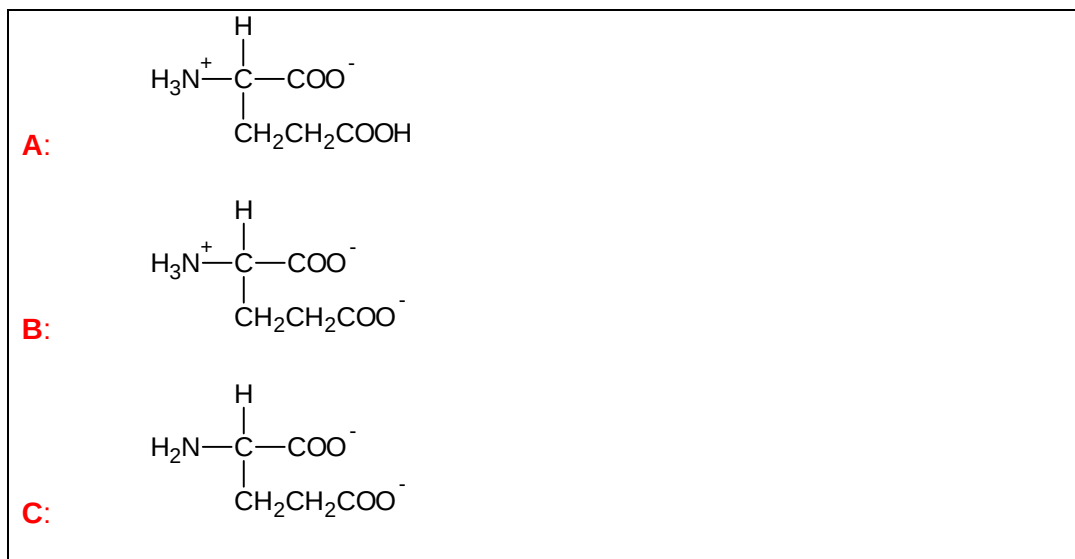
$$\text{p}K_{\text{a}2} = 4.07$$

$$\text{p}K_{\text{a}3} = 9.47$$



(i) Suggest the structures of **A**, **B** and **C**.

[2]



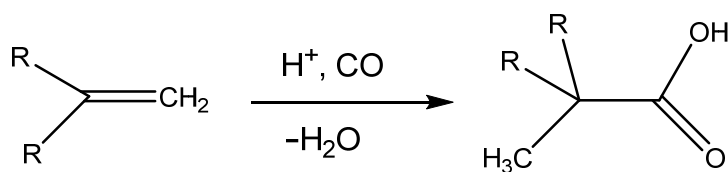
(ii) State the pH of the resultant solution when 1.5 moles of OH^- is added to the protonated glutamic acid.

[1]

Equimolar quantities of A and B are present.

$\text{pH} = \text{pK}_{\text{a}2} = 4.07$

- (d) In the Koch reaction, carboxylic acids are formed from the acid catalysed reaction between alkenes and carbon monoxide.



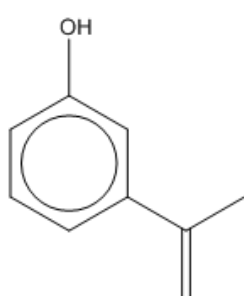
R = alkyl group

- (i) Compound **A** has molecular formula $\text{C}_9\text{H}_{10}\text{O}$. It undergoes the Koch reaction to produce compound **B**.
Compound **A** also decolourises aqueous bromine to produce a white precipitate. When reacted with sodium metal, it produces a gas that extinguishes a lighted splint. It also reacts with acidified potassium manganate (VII), KMnO_4 , to produce compound **C**, $\text{C}_8\text{H}_8\text{O}_2$. When treated with alkaline iodine, compound **C** produces a yellow precipitate. No precipitate was observed when Tollens' reagent was added to compound **C**.

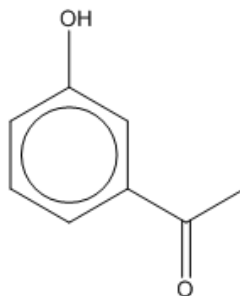
Suggest the structures of compounds **A**, **B** and **C**. Explain all the reactions involved.

[6]

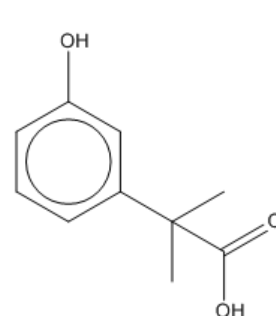
- ✓ High C:H ratio – suggest the presence of an aromatic ring
- ✓ Compound **A** undergoes Koch reaction – suggest presence of a $-\text{C}=\text{C}-$ in compound **A** and a carboxylic acid functional group in compound **B**
- ✓ Compound **A** undergoes **electrophilic substitution** with aqueous bromine to produce a white precipitate – suggest the presence of phenol functional group
- ✓ Compound **A** undergoes **electrophilic addition** with aqueous bromine which gets decolourised – suggest the presence of $-\text{C}=\text{C}-$ functional group
- ✓ Compound **A** undergoes **redox** reaction with Na(s) to produce hydrogen gas – suggest presence of phenol functional group
- ✓ Compound **A** undergoes **oxidation** with acidified KMnO_4 – suggest presence of an alkene
- ✓ Compound **C** undergoes **oxidation** with alkaline iodine – suggest presence of $-\text{CH}_3\text{CO}$ group
- ✓ Compound **C** **does not** undergo **oxidation** with Tollens' reagent – absence of aldehyde FG /presence of a ketone functional group



A

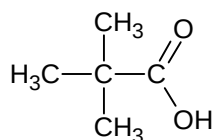


C

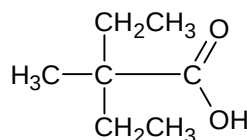


B

(ii) Carboxylic acids, **D** and **E** could be formed using the Koch reaction.



carboxylic acid **D**



carboxylic acid **E**

Carboxylic acid **D** has a lower pK_a value than carboxylic acid **E**. Explain the difference in their pK_a values.

[3]

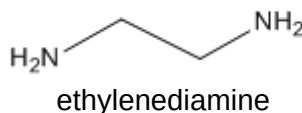
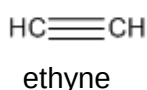
- Carboxylic acid **E** has a bigger alkyl group which is a stronger electron donating group
- which intensifies the negative charge of the conjugate base to a greater extent and destabilises it
- Carboxylic acid **E** dissociates to a less extent and produces less H⁺ ions

[Total: 19]

Section B (20 marks)

Answer **one** question from this section on the writing paper provided.

- 4 Organic molecules such as ethyne, C₂H₂, and ethylenediamine can function as ligands to transition metal ions.



- (a) A blue complex salt **A** has the molecular formula NiN₅H₁₇OCl₂ (*M*_r = 232.7).

1.00 g of **A** reacts completely with 25.00 cm³ of 0.344 mol dm⁻³ silver nitrate solution.

When excess ethylenediamine was added to the blue solid **A**, a violet solution containing complex **B** was produced. **A** and **B** have the same coordination number.

- (i) Calculate the amount of free chloride ions per mole of **A**.

[1]

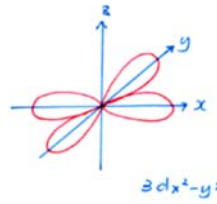
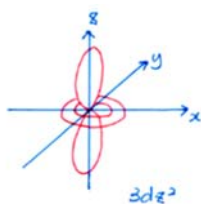
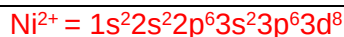
$$\begin{aligned} \text{Amount of A} &= \frac{1}{232.7} \\ &= 4.2974 \times 10^{-3} \text{ mol} \\ \text{Amount of Ag}^+ &= \frac{25}{1000} \times 0.344 \\ &= 8.60 \times 10^{-3} \text{ mol} \\ &= \text{Amount of free Cl}^- \end{aligned}$$

$$\frac{\text{Amount of A}}{\text{Amount of Cl}^-} = \frac{4.2974 \times 10^{-3}}{8.60 \times 10^{-3}} \approx \frac{1}{2}$$

Hence, 1 mol of A contains **2 mol** of free chloride ions.

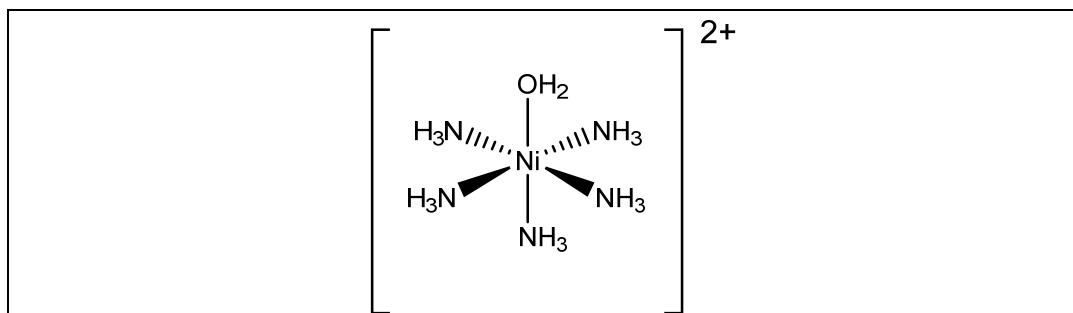
- (ii) State the full electronic configuration of the nickel ion. Draw fully-labelled diagrams of the orbitals which experience the greatest repulsion between the *d* electrons and the lone pair of electrons on the ligands.

[2]



- (iii) With reference to your answer in (a)(i) and (a)(ii), draw a three dimensional diagram to illustrate the shape of the complex ion in A.

[1]



- (iv) Suggest a possible formula of the complex ion in B.

[1]

Any of the following :

- $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)(\text{NH}_3)_4]^{2+}$
- $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)(\text{H}_2\text{O})(\text{NH}_3)_3]^{2+}$
- $[\text{Ni}((\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2(\text{NH}_3)_2]^{2+}$
- $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2(\text{H}_2\text{O})(\text{NH}_3)]^{2+}$
- $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$

- (b) State the hybridisation of the carbon atoms present in ethyne. Hence, in terms of orbital overlap, justify why ethyne can function as a ligand.



ethyne

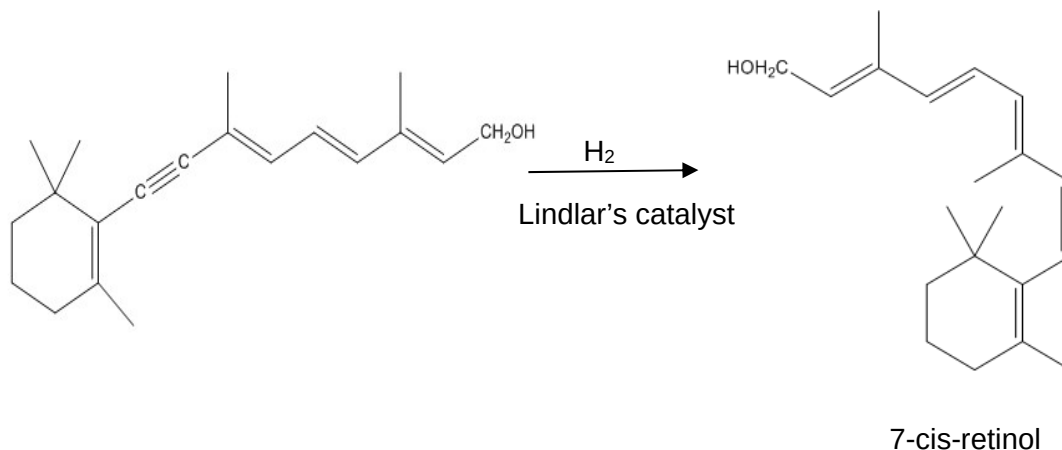
[3]

1. In ethyne, each of the carbon atom is *sp* hybridised.
2. The 2 unhybridised *p* orbital of the two carbon atoms, which is perpendicular to the plane of *sp*² hybrid orbitals, overlaps laterally or sideways to form 2 π bonds.

3. The electron density around the π bond is high, which allows ethyne to function as a ligand (or the pi electrons are not directly involved in holding the 2 carbon nuclei in place and since they are loosely held by the 2 carbon nuclei, they are available for reaction).

- (c) Alkynes are reduced to alkenes, using the Lindlar's catalyst, which are finely divided palladium metal precipitated onto a calcium carbonate support.

An example is in the synthesis of 7-cis-retinol.



- (i) State the number of cis-trans isomers present in a molecule of 7-cis-retinol.

[1]

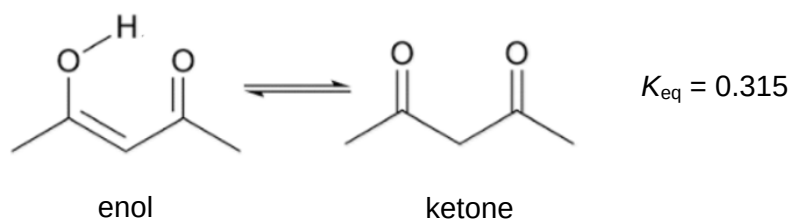
$$2^4 = 16$$

- (ii) Describe the mode of action by the Lindlar's catalyst.

[3]

- ✓ Palladium functions as a heterogeneous catalyst.
- ✓ Availability of partially filled d orbitals in palladium metal allows reactant molecules to adsorb onto metal surface.
- ✓ This weakens the bonds within the reactants.
- ✓ Reactant molecules are also brought closer together, increasing surface concentration on the catalyst.
- ✓ As such, activation energy is lowered.
- ✓ Once the products are formed, the product molecules are desorbed from the metal surface.

- (d) Alkyne can also undergo hydration to form an enol which rapidly rearranges itself into a ketone. Such a rearrangement is known as ketone tautomerisation. An example is that of 2,4-pentanedione.

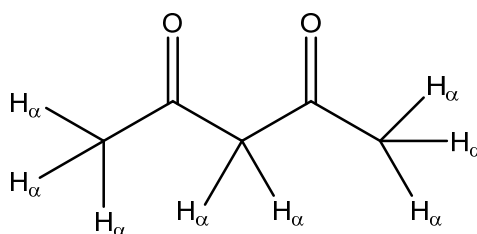


Using the data given, deduce if the tautomerisation reaction is spontaneous.

[2]

$K < 1$, suggest that position of equilibrium lies to the left.
 $\Delta G > 0$, Not spontaneous

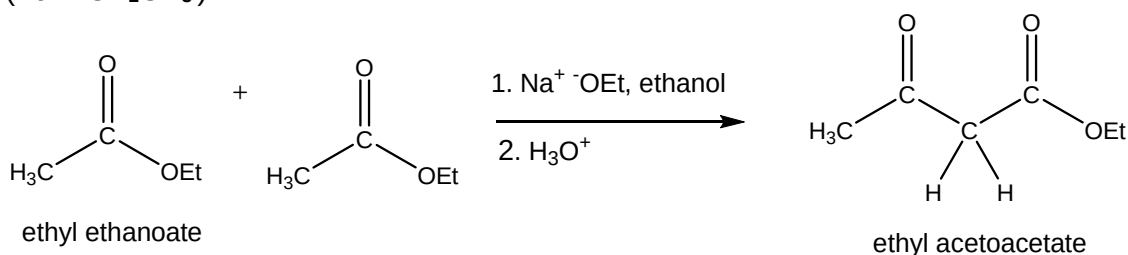
- (e) Carbonyl compounds, like 2,4-pentanedione contains acidic α hydrogen atoms.



Esters, like ketones and aldehydes, can also contain acidic α hydrogen atoms. When deprotonated by a suitable base, a carbonyl condensation reaction can occur.

An example is the formation of ethyl acetoacetate from ethyl ethanoate.

(Et : $-\text{CH}_2\text{CH}_3$)



Part of the mechanism is described as follows:

Step 1:

A base, ethoxide ion (^-OEt), abstracts an acidic alpha hydrogen atom from the ethyl ethanoate molecule, yielding a RCH_2^- nucleophile.

Step 2:

The resultant RCH_2^- nucleophile adds to a second ester molecule, giving a tetrahedral alkoxide intermediate:

Step 3:

The tetrahedral intermediate expels the ethoxide ion to yield the new carbonyl compound, ethyl acetoacetate.

- (i) Suggest the role of ethanol.

[1]

Organic solvent

- (ii) Suggest, with a reason, why the alpha hydrogen of the ester, ethyl ethanoate in Step 1 is acidic.

[1]

The conjugate base produced is stabilised by the electron withdrawing ester functional group (or by resonance)

- (iii) Using the information given, suggest the mechanism for the reaction. Show relevant lone pairs, dipoles and charges, and indicate the movement of electron pairs with curly arrows.

[3]

A base, ethoxide ion (OEt^-), abstracts an acidic alpha hydrogen atom from the ethyl ethanoate molecule, yielding a RCH_2^- nucleophile.

+ EtOH

Step 2:

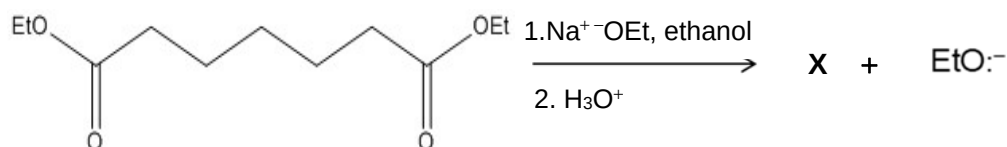
The resultant RCH_2^- nucleophile adds to a second ester molecule, giving a tetrahedral alkoxide intermediate:

Step 3:

The tetrahedral intermediate expels the ethoxide ion to yield the new carbonyl compound, ethyl acetoacetate.



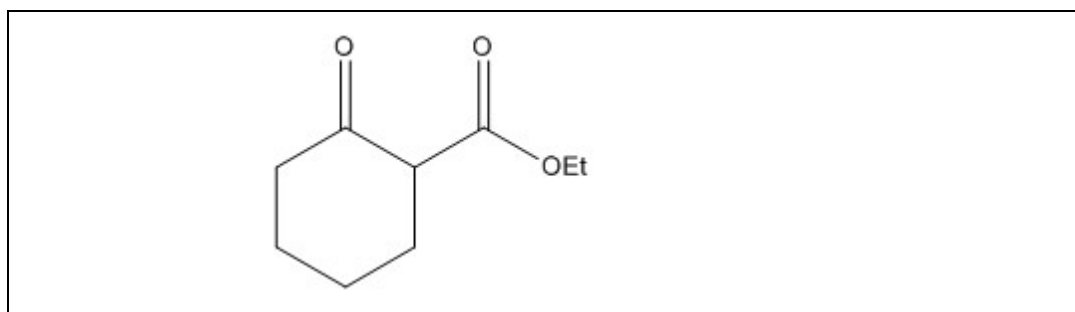
(iv) Compound **X** can be synthesised from diethyl heptanedioate via the carbonyl condensation reaction.



diethyl heptanedioate

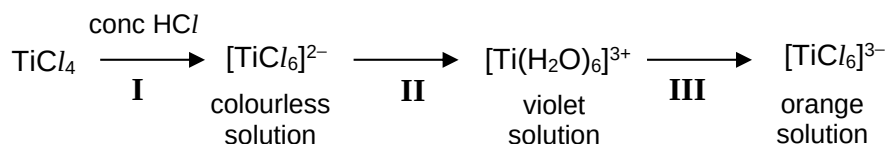
Suggest the structure of compound **X**.

[1]



[Total: 20 m]

5 (a) The following shows a series of reactions involving titanium compounds.



(i) Suggest the type of reaction for **II** and **III**.

[2]

II: reduction or redox

III: ligand exchange

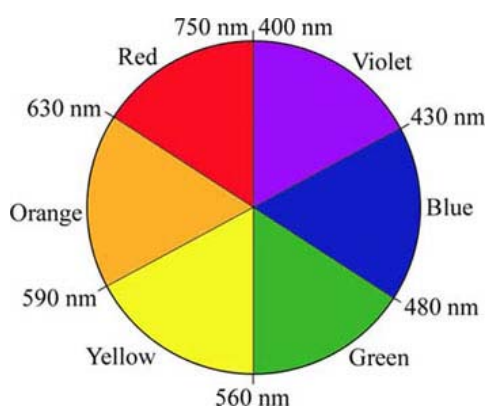
- (ii) Explain why a solution of $[\text{TiCl}_6]^{2-}$ is colourless while that of $[\text{TiCl}_6]^{3-}$ is orange. [2]

Ti^{4+} in $[\text{TiCl}_6]^{2-}$ has empty d orbitals while Ti^{3+} in $[\text{TiCl}_6]^{3-}$ contains partially filled d orbitals.

Hence electron can be promoted from lower energy d orbital to the higher energy d orbital ($d-d$ transition) for $[\text{TiCl}_6]^{3-}$ but not for $[\text{TiCl}_6]^{2-}$.

- (iii) White light contains all the colours in the visible spectrum, and each of these colours is associated with a certain wavelength λ . The figure below shows a colour wheel with approximate wavelength values in nanometres for different colour light.

The formula relating energy gap between d orbitals, ΔE and wavelength λ is given as $\Delta E = \frac{hc}{\lambda}$, where h is the Planck's constant and c is the speed of light. As wavelength decreases, the energy gap increases.



Using the information given and those in (a), deduce whether $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ or $[\text{TiCl}_6]^{3-}$ has a bigger energy gap between the d orbitals. [1]

Violet $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ absorbs in the yellow region while orange $[\text{TiCl}_6]^{3-}$ absorbs in the blue region. Blue colour has a shorter wavelength than yellow and thus indicates a bigger energy gap between the 2 sets of orbitals.

- (iv) Explain why titanium forms complexes with different oxidation states while calcium is unable to do so. [1]

Ti can show variable oxidation states as the energy difference between the $3d$ and $4s$ orbitals of titanium is relatively small. With sufficient energy, the $4s$ and inner $3d$ electrons can be removed.

- (b) Organotitanium compounds such as CH_3TiCl_3 can function as nucleophiles. The methyl group, CH_3^- , acts as a nucleophile.

The reduction reaction of an ester, methyl propanoate, $\text{C}_2\text{H}_5\text{COOCH}_3$ with CH_3TiCl_3 , followed by acidification produces a tertiary alcohol, $\text{C}_2\text{H}_5\text{C}(\text{CH}_3)_2\text{OH}$, as shown.



- (i) The mechanism is proposed as follows.

Step 1:

The nucleophile CH_3^- from CH_3TiCl_3 adds to the carbonyl carbon of the ester to form a tetrahedral alkoxide intermediate.

Step 2:

A ketone and a methoxide ion, CH_3O^- are produced from the alkoxide intermediate.

Step 3:

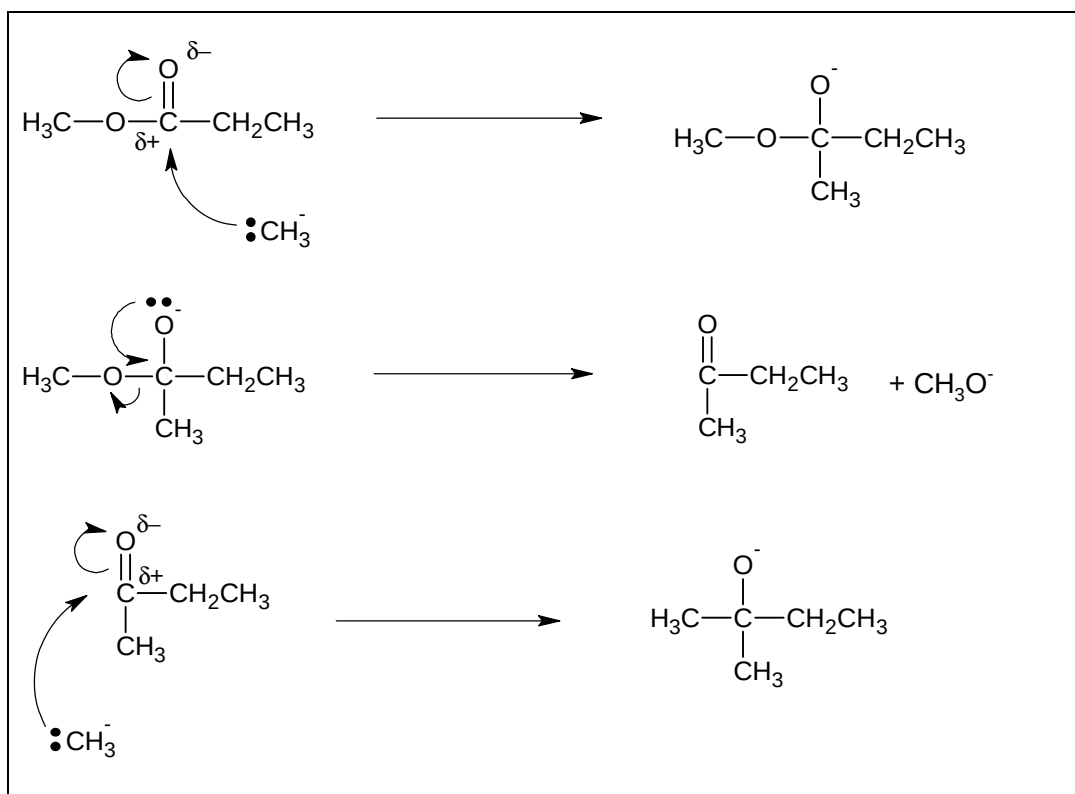
Another CH_3^- from CH_3TiCl_3 adds to the carbonyl carbon of the ketone to form another tetrahedral alkoxide intermediate.

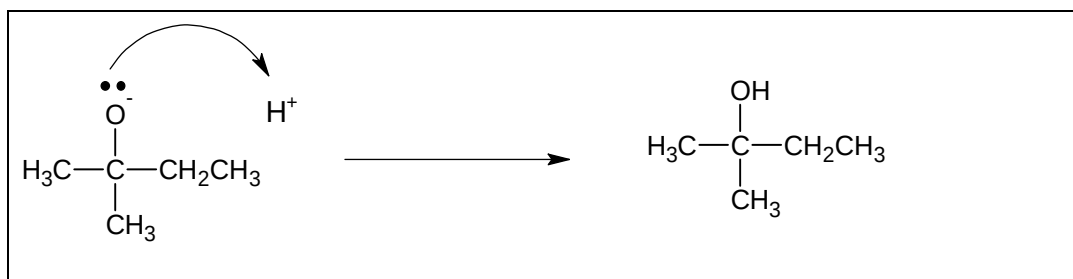
Step 4:

Protonation of the alkoxide intermediate forms the product $\text{C}_2\text{H}_5\text{C}(\text{CH}_3)_2\text{OH}$.

Using the information given, suggest the mechanism for the reaction. Show relevant lone pairs, dipoles and charges, and indicate the movement of electron pairs with curly arrows.

[4]





- (ii) Amides can also be reduced by organotitanium compounds to form the corresponding alcohol and amine. However, this reaction occurs at a slower rate as compared to that of an ester.

Suggest an explanation for this observation.

[1]

Oxygen is more electronegative than nitrogen, hence the carbonyl carbon of the ester is more electron deficient compared to that of the amide. Hence the ester is more susceptible to nucleophilic attack by the reducing agent.

- (iii) Suggest a simple chemical test to distinguish between $\text{C}_2\text{H}_5\text{COOCH}_3$ and $\text{C}_2\text{H}_5\text{C}(\text{CH}_3)_2\text{OH}$.

[2]

test: anhydrous PCl_5

observations: white fumes of HCl observed for $\text{C}_2\text{H}_5\text{C}(\text{CH}_3)_2\text{OH}$ and no fumes observed for $\text{C}_2\text{H}_5\text{COOCH}_3$

Other alternatives are acceptable eg Na test.

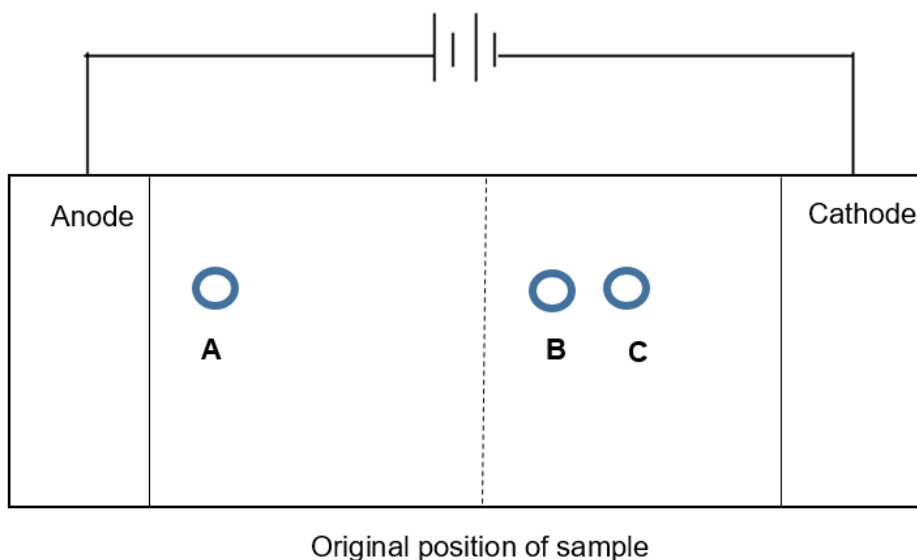
- (c) The following amino acids are some of the amino acids present in the primary sequence of haemoglobin.

amino acid	formula of side chain (R' in $\text{R}'\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$)	isoelectric point
valine	$-\text{CH}(\text{CH}_3)_2$	6.00
glutamic acid	$-\text{CH}_2\text{CH}_2\text{COOH}$	3.15
asparagine	$-\text{CH}_2\text{CONH}_2$	5.41

At an intermediate pH, called the isoelectric point (pI), the amino acids will be *zwitterionic* and have *no* net charge.

An electrophoresis experiment is run on a solution containing the above three amino acids at pH 5.00. The relative positions of the amino acids are shown in the diagram below.

The rate at which the amino acids move through the gel is inversely proportional to its mass.



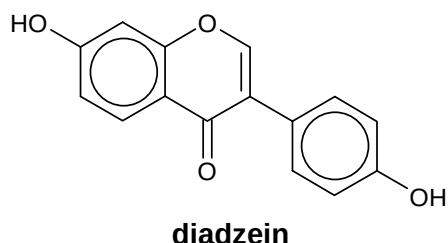
Identify the structures **A**, **B** and **C**. Hence, explain the position of **B** relative to **C**.

[2]

A: Glutamic acid; **B**: Asparagine ; **C** : Valine

Asparagine has a larger molecular mass than valine. Since, the rate in which the amino acids move through the gel is inversely proportional to its mass, asparagine will migrate to the cathode at a slower rate.

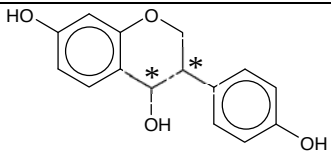
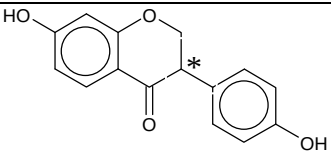
- (d) Soy beans, and especially the dofu made from them, are a good source of dietary isoflavenoids, which are claimed to help in the prevention of some cancers. The major isoflavenoid in soy is diadzein.



When diadzein is treated with H_2 and Ni , compound **D**, $\text{C}_{15}\text{H}_{14}\text{O}_4$, is formed. One mole of compound **D** reacts with three moles of sodium metal. **D** also dissolves in $\text{NaOH}(\text{aq})$. **D** reacts with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to give compound **E**, $\text{C}_{15}\text{H}_{12}\text{O}_4$, which gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent.

Suggest a structural formula for **D** and for **E**, identifying any chiral carbon atoms. Explain the reactions which occur.

[5]

D	E
	

Structures of **D** and **E** with chiral carbon correctly identified

The alkene and ketone in diadzein undergoes reduction with H_2 and Ni to form **D**.

1 mole of **D** undergoes redox with 3 moles of Na as it has 3 $-OH$ groups.

D dissolves in NaOH(aq) as it contains acidic phenol and undergoes acid-base reaction.

The secondary alcohol in **D** is oxidised to ketone.

E contains a ketone group thus it gives an orange precipitate with 2,4-DNPH via condensation reaction.

[Total: 20]

END OF PAPER